

The component or the entity that makes the core of the FreeBSD environment and operating system is called the kernel. It acts as the fundamental interface between a system's hardware and its different processes. Subsequently, its functionality is not only confined to communicating between the two but also managing available resources in the best and most efficient manner possible. The kernel in the FreeBSD operating system is monolithic and modular in nature. It serves many purposes such as administering security controls, disk access, managing memory, networking, and the like.

Even though the configuration of a large part of the FreeBSD environment can be carried out dynamically, it is advisable, and rather necessary at times, to compile and configure a custom kernel. In a broader sense of things, the kernel in the FreeBSD operating system can essentially be visualised as a resource manager that acts as an abstract layer for all the applications that are to be serviced and executed. It is the site where the management of hardware, processes and other low-level details occurs.

Need for a custom kernel

Initially, the FreeBSD environment used a kernel that was monolithic in nature. It was in the form of a single large program and only lent support to a predetermined roster of devices. To effect any changes in the kernel's behaviour, one had to first compile and then subsequently restart a different and new kernel altogether. In the current scheme of things, a major part of the functionality and utility of the kernel within a FreeBSD environment stands revamped to the architecture of modules. Consequently, this allows for the features to either be dynamically loaded or unloaded from the kernel as and when deemed necessary. Additionally, this also enables the kernel to adjust almost instantaneously to the new hardware, opening up the scope of introducing new functionality to the kernel in the process. Thus, the current model of the FreeBSD kernel is classified to be modular in nature.

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root@srvr02: ~# ssh -o 'UserKnownHostsFile=/dev/null' -o 'StrictHostKeyChecking=no' https://svn0.us-east.freebsd.org/
Error validating server certificate for 'https://svn0.us-east.freebsd.org:443':
 - The certificate is not issued by a trusted authority. Use the
   fingerprint to validate the certificate manually!
Certificate information:
 - Hostname: svn0ir.gsv.freebsd.org
 - Valid: from Jul 29 22:01:21 2013 GMT until Dec 13 22:01:21 2040 GMT
 - Issuer: svn0ir.gsv.freebsd.org, clusteradm, FreeBSD.org, CA, US(clusteradm@f
  reebsd.org)
 - Fingerprint: 1C:BD:85:95:11:9F:EB:75:A5:4B:C8:A3:FE:08:E4:02:73:06:1E:61
(R)ject, accept (t)emporarily or accept (p)ermanently? █
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Image Source: explorelinux.com